

# Lakshitha Eranga

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## Summary

Computer Science graduate with research focused on the intersection of systems engineering and artificial intelligence. Experienced as a systems architect designing scalable distributed infrastructure and optimizing complex AI workloads in production environments. Primary research interests lie in investigating how concurrency, scheduling, and memory behavior shape the performance and cost of AI infrastructure.

## Education

**University of Peradeniya**, BSc (Hons) in Computer Science (4-Year Degree) Jun 2021 – Dec 2025

- GPA: 3.74/4.0 (First Class Honors)
- Minors:** Mathematics, Statistics. **Specialty Areas:** Distributed Systems Architecture, Parallel Processing, Cloud Infrastructure & DevOps, AI Workload Optimization.

## Research Achievements

**ISBM – Bangkok 2026** (Conference Paper) *Presented (Virtual)*

- Parallelism Isn't Free: Two Hidden Walls in a Heterogeneous Media-Understanding Pipeline*
- Proceedings to be published in *Springer – Lecture Notes in Networks and Systems*

**Ceylon Journal of Science** (Journal Paper) *In Review*

- A two-stage output-space parameter-efficient transfer learning framework for low-resource medical imaging adaptation*

**Research Café 2026**, University of Peradeniya *Poster Presented*

- Presented a poster on the undergraduate research into parameter-efficient transfer learning for low-resource medical imaging.

## Honors and Awards

**University Award for Academic Excellence**, University of Peradeniya 2026

- Recognized for maintaining outstanding academic performance within the Department of Statistics and Computer Science.

## Research Experience

**Research Collaborator (Voluntary)**, Dr. Erunika Dayaratna's Research Group, University of Peradeniya

### Heterogeneous Media-Understanding Pipeline

- Published:** ISBM 2026 paper on memory and concurrency bottlenecks in heterogeneous inference pipelines.
- Ongoing:** Reading and reviewing current literature on memory and dynamic scheduling for heterogeneous AI workloads, continuing from the published work.

### Infrastructure Optimization for Edge AI Deployment in Semi-Automated Electric Vehicles over Public 5G/6G Networks

- Ongoing:** Collaborating on the infrastructure layer, building on the group's prior work on edge-AI vehicle systems.

**Undergraduate Research – Transfer Learning in Low-Resource Medical Imaging** Undergraduate Research  
Supervisor: Dr. Ruwanthini Siyambalapitiya

- Reviewed existing pretrained models and fine-tuning strategies, including Vision Transformers (ViTs) and CNNs.
- Explored novel fine-tuning techniques to independently enhance model performance in low-resource medical imaging scenarios.
- Investigated multi-modal fine-tuning approaches for better generalization across various medical datasets.

## Work Experience

**Temporary Demonstrator**, University of Peradeniya Jun 2026 – Present

- Teach practical laboratory courses and lead hands-on technical sessions for undergraduate students in the Department of Statistics and Computer Science.
- Evaluate student coursework, design practical assignments, and foster foundational skills in core computer science modules.

**Backend & Infrastructure Architect (Contract)**, Social Generation Inc. – USA (Remote)

Jun 2025 – Jan 2026 (Part-Time) · Jan 2026 – Jun 2026 (Full-Time)

- **AI Marketing Video Pipeline:** Architected a highly scalable, autoscaling backend pipeline that analyzes video inputs and autonomously rearranges them into marketing assets, achieving a 3.5x speed boost in task execution time to support rapid user growth.
- **Heterogeneous AI Integration:** Orchestrated a distributed hybrid environment, integrating local CPU-bound models for transcription processing with external AI APIs for heavy video analysis.
- **Asynchronous Processing:** Built background job pipelines using Celery and message brokers to process heavy AI workloads without slowing down the main API.
- **Cloud Infrastructure:** Managed scalable cloud environments using Terraform and built CI/CD pipelines for automated testing and zero-downtime deployments.

**AI and Automation Engineering Intern**, Taino Spain S.L – Spain (Remote)

Sep 2025 – Dec 2025

- **RAG Systems:** Engineered a LangChain-based RAG chatbot with vector databases to handle customer inquiries via WhatsApp and web.
- **Workflow Automation:** Automated social media content generation and scheduling pipelines using n8n and Gemini models, drastically reducing manual workload.

## Related Courses

Distributed Systems, Parallel Processing, Server-side Programming, Advanced Networking, Machine Learning, Neural Networks and Deep Learning, Database Management Systems, Data Structures and Algorithms.

## Skills

- **Languages:** C, C++, Python, Java, JavaScript, TypeScript, SQL
- **Distributed & Parallel Computing:** OpenMP, MPI, CUDA, Celery, Redis
- **Cloud, DevOps & Systems:** Linux, AWS, Terraform, Docker, Dokploy, FFmpeg, HLS Video Streaming
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn
- **Web Frameworks & Databases:** Django, Node.js, NestJS, FastAPI · PostgreSQL, MongoDB, MySQL

### Certifications:

- Google Cybersecurity Professional Certificate, Coursera (2023)
- Deep Learning Specialization, Coursera (2021)
- Introduction to Big Data, Coursera – UC San Diego (2022)

## Professional Organizations and Affiliations

**President – Computer Society, University of Peradeniya (CSUP)**

2024–2025

- Led the Computer Society, overseeing project coordination, member collaboration, and event planning.

**Organizing Committee Member – Research Congress (RESCON)**, Postgraduate Institute of Science, University of Peradeniya

2026

- Supported event logistics and academic presentations for the research congress.

**Organizing Committee Member – Science Undergraduate Research Symposium (SURS)**, University of Peradeniya

2024

- Supported event logistics and academic presentations for the undergraduate research symposium.

## Extra Curricular Activities

**Hackathons & Competitions**

2024

- **Manthra 2024 CTF, IEEE NSBM** – Secured 6th place out of competing university teams.
- **Data Science Mini Hackathon 2024, UOC Statcircle** – Top 15 ranking.

**Independent Tech Writer – Medium**

*lakshithe.medium.com*

Present

- Independently research and author articles on technology, personal development, and AI/ML topics.

## References

**Dr. Ruwanthini Siyambalapitiya**

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**Dr. Erunika Dayaratna**

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